INCH-POUND
MIL-PRF-21480/25(AS)
23 April 2020

PERFORMANCE SPECIFICATION SHEET

GENERATOR SYSTEM, ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS (H-1 AIRCRAFT)

This specification has been approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-21480.

SCOPE:

1. Generator class. The generator is classified as a class W, as modified herein.

2. Part or identifying number (PIN). Components that make up the generator systems covered by this specification sheet are identified using the following PIN:

Component Name	Part Number	Figure Number
Generator	M21480/25-1	See figure 1
Generator Control Unit (GCU)	M21480/25-2	See figure 2
Current Transformer	M21480/25-3	Not Applicable

REQUIREMENTS:

1. Dimensions and weight. The dimensions and weight of the generating system components shall be as shown on figures 1 through 3 of this specification sheet.

2. Capacity. The rating of the generating system shall be not less than 20/23.8 kVA. See 3.4.8 and sub-paragraphs as modified herein.

3. Generator system characteristics. The generator system characteristics shall be as shown on figure 1 of this specification sheet.

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4. MIL-PRF-21480 variance. The generating system shall comply with MIL-PRF-21480, except as follows:

4.1 The following paragraphs of MIL-PRF-21480 are not applicable:

3.4.2.10 Chemical, biological, and radiological (CBR).
3.4.2.11 Lightning, electromagnetic pulse, and directed energy weapons.
3.4.8.2.1 Motor starting (AC power).
3.4.8.2.2 Emergency overload (AC power).
3.4.9.6.2 Liquid cooling, and subparagraphs a through e.
3.5.7.4.a Frequency protection.
3.5.7.10 Arc fault power feeder protection.
3.5.9 Paralleling.
3.5.10 Generator output characteristics.

4.2 Modify the following paragraphs of MIL-PRF-21480:

3.3.2 Moisture resistance. Add the following to the end of the paragraph: “Externally mounted connector receptacles shall be environment resistant conforming to SAE AS31021 (class E or class R), or SAE AS34521 (class W).”

3.4.2.1.1 Blast air-cooled and integral fan-cooled components. Delete the last sentence and substitute: “The altitude limit for continuous operation is 15,000 feet.”

3.4.2.1.2 Liquid-cooled and self-cooled components. Delete the last sentence and substitute: “The altitude limit for continuous operation is 15,000 feet.”

3.4.2.8 Acceleration forces. Delete last sentence.

3.4.6 Consolidation of generating system components. Delete the last sentence and substitute: “The generator control unit shall be a separate component from the generator package (see 6.8.9).”

3.4.7.1 Manual system control. Delete the first sentence and substitute: “Manual system control of the AC power source shall be provided by a two-position switch, type I (see 6.8.2a).”

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3.4.8 System capacity (power source rating). Delete the paragraph in its entirety and replace with the following: "The system AC power source rating shall be based on 115/200 volts, 400 Hz, three-phase, four wire electric power. The direct current (DC) power source shall conform to the requirements of 3.4.8.4.2. The AC and DC power sources shall deliver rated capacity within the generator speed range (class W). All ratings specified herein shall be available at the Point of Regulation (POR) as defined in 6.8.16 with a feeder voltage drop of 0 to 5 volts from the AC output terminals to the POR. Generator three phase load current measurements shall be measured at the load end of the AC generator feeders POR.
The GCU shall provide output measurements of AC three phase load current in form of output ARINC-429 signals. The AC three phase load measurements provided by the GCU shall have a worst case error of 3 percent when actual root mean square (rms) current is less than 50 A and 6 percent error of actual rms current when current is between 50 A and 300 A.

The feeders from the generator output terminals to the line contactor shall be 6 AWG, 6.5±0.5 feet in length. Air cooled generators with a self-cooled rating and a blast-cooled rating shall provide the output capacity specified for each operating mode as required herein (see 3.4.9.6.1)."

3.4.8.4.2 a. Continuous DC capacity. Delete in entirety and replace with: "The DC power source shall be capable of delivering continuous DC current from no load to full rated DC load. The DC source's rated output current capacity for continuous operation shall be not less than 2 amps."

3.4.9.2 Mounting and driving provisions. Delete in entirety and replace with: "The generator shall be attached and secured to the drive pad with a V-band clamp. The V-band clamp shall be designed so that it can be installed or removed by tightening or loosening one self-locking integrally retained fastener. Generator drive pad interface features and dimensions shall be as required on figure 1 of this specification sheet. Provisions for positive alignment during generator package installation shall be as required on figure 1 of this specification. The generator housing shall not rotate when mounted on the drive pad. An anti-rotation pin on the drive pad interface is one acceptable means of preventing rotation."

3.4.9.2.2 Shear section. Delete the second sentence and replace with: "Shaft shear shall occur when applied torque is within 830 to 940 inch-pounds, and shall withstand an acceleration of 2000 RPM per second."

3.4.9.3 Electric connections. Delete in entirety and replace with: "Generator shall be supplied with terminals as shown on figure 1 of this specification sheet. All generator terminal studs shall be stainless steel with minimum hardness of Rockwell C-34. Terminal block designs shall incorporate insulation barriers between terminal connections with clearance to accommodate installation of SAE AS25036 wiring lugs as specified."

3.4.9.4 Direction of rotation. Delete in entirety and replace with: "The direction of rotation, when viewing the generator drive end shall be counter-clockwise. A permanent marking on the generator shall indicate direction of rotation and shall be visible when the generator is mounted."

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3.4.9.5 Efficiency (heat rejection). Delete in entirety and replace with: “Under the test conditions of 4.5.9, generator system efficiency at nominal speed with full rated load (0.75 lagging power factor) shall be not less than 85 percent.”

3.4.11 Color. Change “system components” to “generator and generator control unit.”

3.5.7.3 Undervoltage protection. Delete in entirety and replace with: “The undervoltage protection device shall function to disconnect the system from the load bus when any one phase drops to a value of 95-105 volts for a time delay of 4±1 seconds. The system shall be capable of being reset when all three phases are above drop-out level and before any phase exceeds 105 volts.”

3.5.7.4.b Class W. Delete in entirety and replace with: “The underfrequency protection device shall function to disconnect the system from the load bus when the frequency drops to a value of 370-380Hz for a time delay of 1±0.5 second. Pickup value shall be greater than drop-out level and before the frequency exceeds 380Hz. The minimum difference between the drop-out frequency and pickup frequency level shall be 4Hz.”

3.11 Maintainability. Delete in entirety and replace with: “The generator shall not require any scheduled maintenance other than the minimum 4,000-hour grease lubricated bearing service life requirement of 3.4.9.7. The GCU and CT shall not require any scheduled maintenance.”

4. VERIFICATION

4.5.3.1b Electrical performance at standard temperature. Delete in entirety and replace with: “Measure output voltage and current (generator AC and GCU DC) under each test condition in table II to determine conformance to the requirements of 3.4.8, 3.4.8.1, 3.4.8.2, 3.5.1.1 and 3.5.1.2. Under each of the table II test conditions for continuous operating mode, apply the AC load suddenly and measure the transient response of the AC and DC voltages. After application of these AC loads, measure the steady state regulation of the AC and DC voltages. After component temperatures have stabilized within 4.4.8.2 criteria, remove the AC load suddenly and measure the transient response of the AC and DC voltages. Under the five-minute and five-second AC overload conditions in table II, apply AC overloads suddenly and measure the transient response and regulation of the AC and DC voltages. After the specified period (five minutes or five seconds), remove the AC overload suddenly and measure the transient response of the AC voltages and DC voltage. Under continuous load and five-minute overload conditions in table II, measure steady state AC voltage modulation and waveform (crest factor and distortion) to determine conformance to the requirements of 3.5.2 and 3.5.4. During steady-state conditions, measure DC voltage (ripple amplitude and distortion spectrum) to determine conformance to the requirements of 3.5.5.”

4.5.10 Short circuit capacity (AC and DC). Delete last sentence.

4.5.12 Electromagnetic emissions and susceptibility. Delete third sentence.

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4.5.20.1 Salt-fog. Add the following at the end of the paragraph: “There shall be no evidence of: (1) material degradation such as corrosion; or (2) blistering of paint or degradation of electrical insulation.”

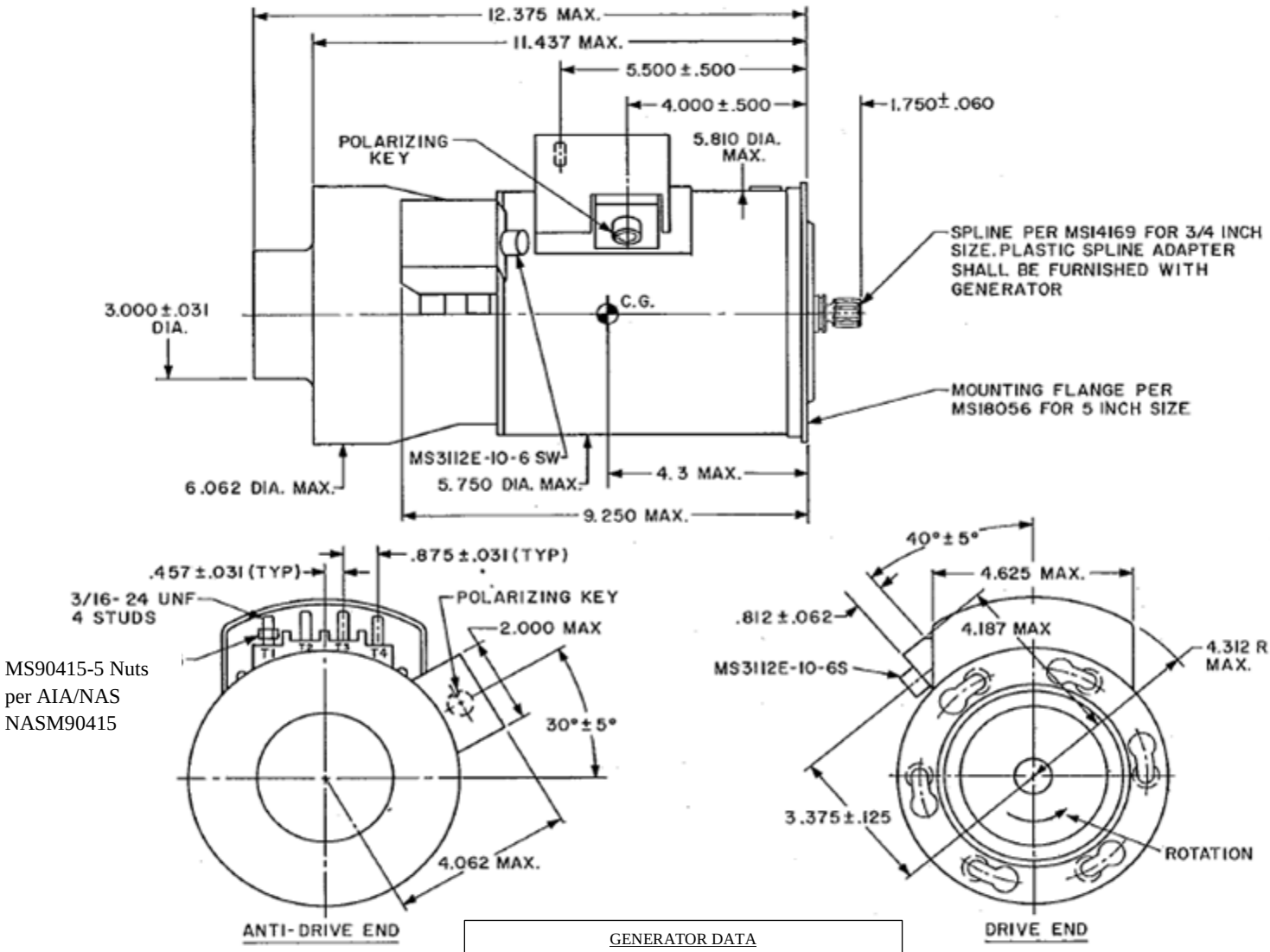
4.5.21 Fungus. Add the following at the end of the paragraph: “There shall be no evidence of: (1) direct attack of fungus on the generator system components such as material breaking down due to fungus growth; or (2) damage to fungus-resistant material due to indirect attack. See 2.1.1.1 of MIL-STD-810, test method 508.”

4.5.22 Sand and dust. Add the following at the end of the paragraph: “There shall be no evidence of: (1) binding, clogging, seizure or blocking of moving parts; or (2) compromising of protective coatings and seals. See 5.1 of MIL-STD-810, test method 510.”

4.5.23 Vibration. Add the following at the end of the paragraph: “The GCU and CTs shall be subjected to vibration levels and durations for category 4.”

4.5.35 Acceleration (inertia loads). Delete in entirety and replace with: “All systems components shall be subjected to the acceleration tests of MIL-STD-810, method 513, procedure II. The required acceleration level for all directions in procedure II (Operational test) shall be 6.0g. The 6.0g shall be applied along the mutually perpendicular axes in two opposite directions along each axes. The test time duration in each axis shall be at least one minute following centrifugal stabilization and the system components shall be operated at continuously at full load during the test.”

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MS90415-5 Nuts per AIA/NAS NASM90415

GENERATOR DATA	
RATED VOLTAGE	120/208 VOLTS
RATED OUTPUT	20 KVA
PHASE	3
FREQUENCY	380 TO 420 Hz
SPEED RANGE	11,400 TO 12,600 RPM
MAX. SPEED FOR REGULATION	13,000 RPM
OVERSPEED	15,000 RPM
RATED POWER FACTOR	.75 LAG TO 1.0
EFFICIENCY MIN AT RATED LOAD	85%
MAXIMUM WEIGHT	23.5 POUNDS
OVERHUNG MOMENT	125 IN.LBS (MAX)
SHAFT SHEAR SECTION	830-940IN LBS.
SELF COOLING	YES
BEARING GREASE	MIL-PRF-81322

FIGURE 1. Generator, P/N M21480/25-1.

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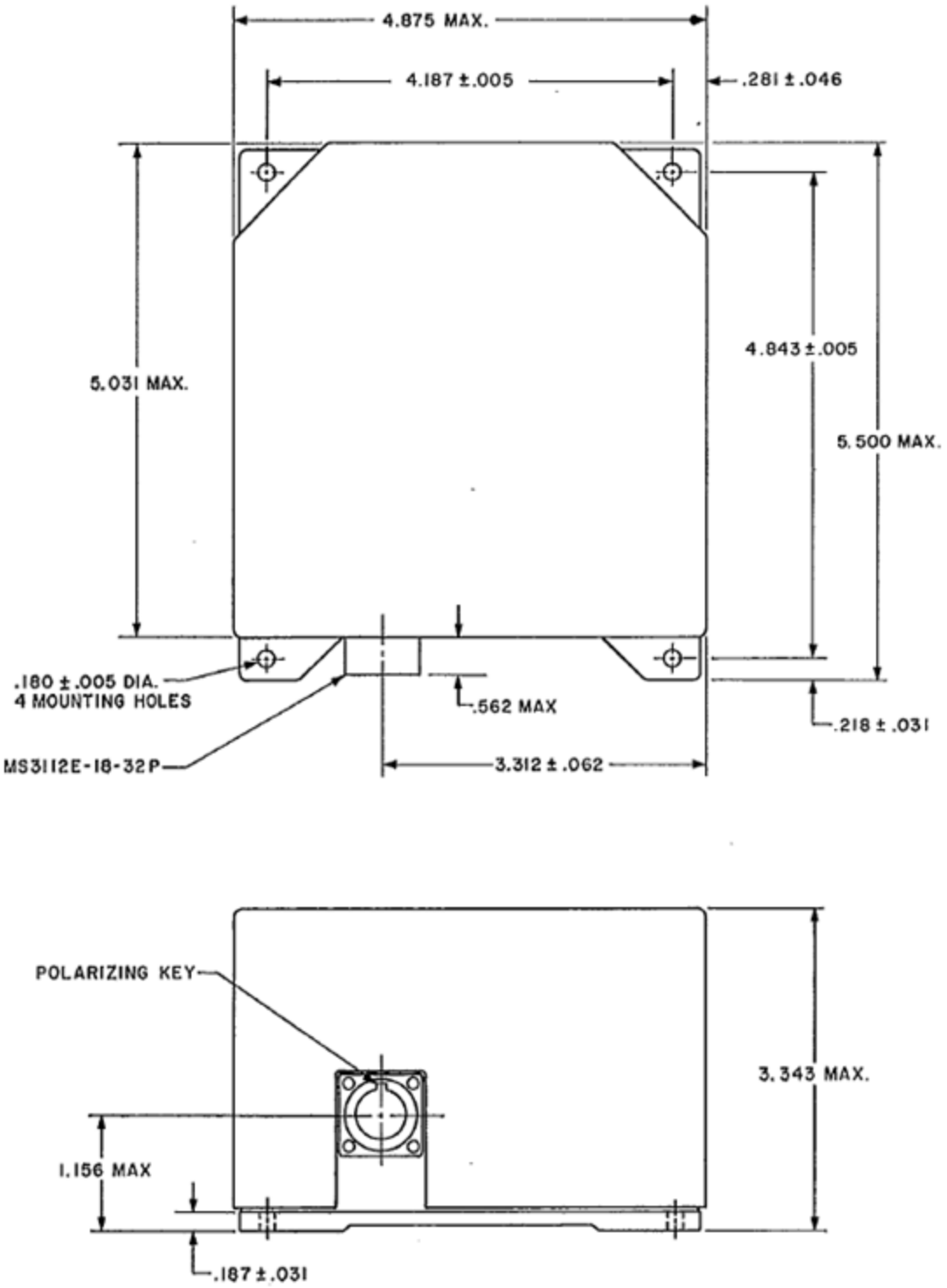


FIGURE 2. Generator control unit, weight 3 lb. max, P/N M21480/25-2.

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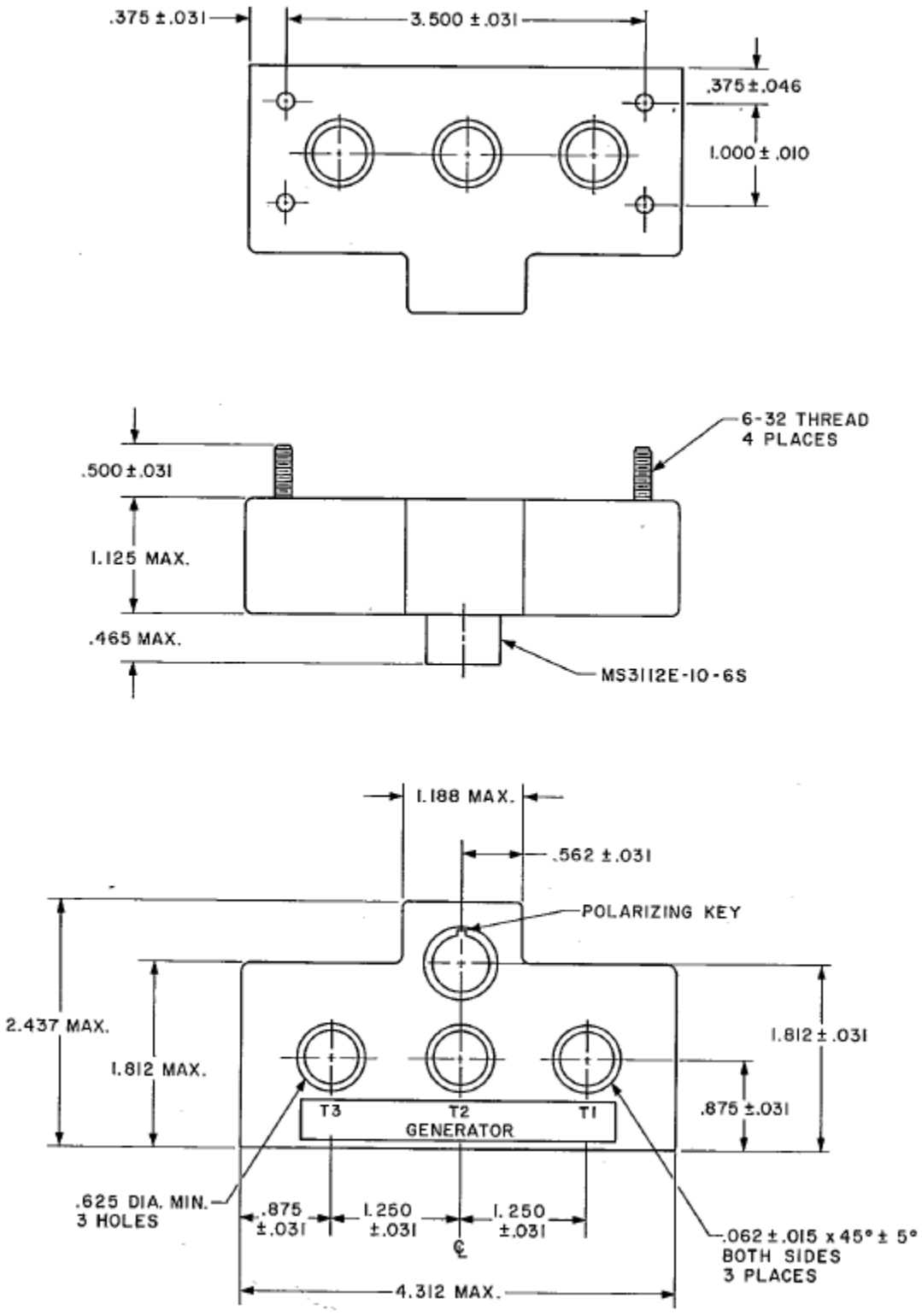


FIGURE 3. Current transformer, weight 0.75 lb. max, P/N M21480/25-3.

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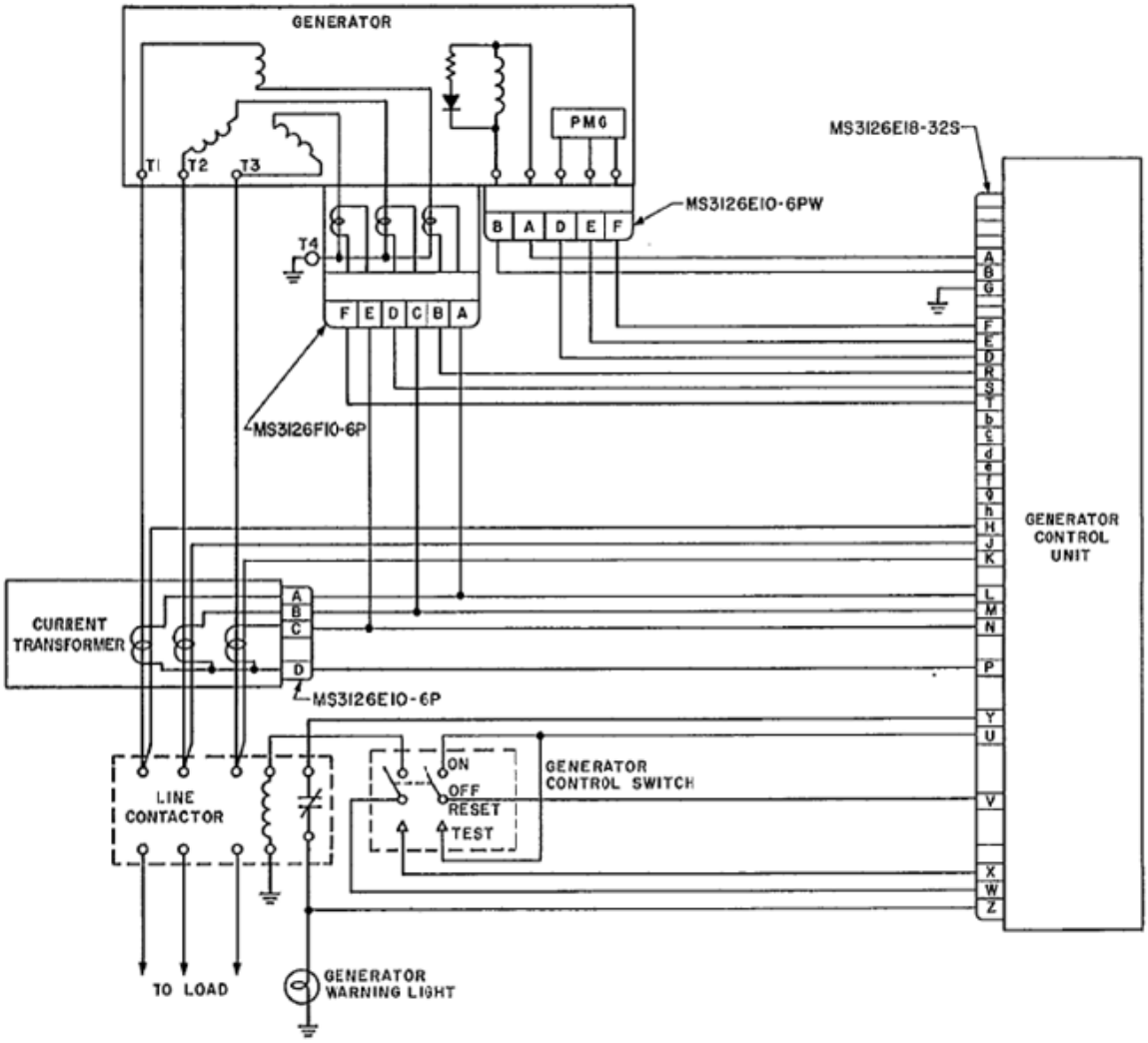


FIGURE 4. Test circuit.

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Referenced documents. In addition to MIL-PRF-21480, this document references the following:

MIL-STD-810	MS3112	MS3126
MS14169	MS18056	SAE AS25036
SAE AS31021	SAE AS34521	MIL-PRF-81322
AIA/NAS NASM90415		

Custodian:
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